

Definition 18.4.4. The *conditional expectation* $\text{Ex}[R \mid A]$ of a random variable R given event A is:

$$\text{Ex}[R \mid A] ::= \sum_{r \in \text{range}(R)} r \cdot \Pr[R = r \mid A]. \quad (18.4)$$

For example, we can compute the expected value of a roll of a fair die, given that the number rolled is at least 4. We do this by letting R be the outcome of a roll of the die. Then by equation (18.4),

$$\text{Ex}[R \mid R \geq 4] = \sum_{i=1}^6 i \cdot \Pr[R = i \mid R \geq 4] = 1 \cdot 0 + 2 \cdot 0 + 3 \cdot 0 + 4 \cdot \frac{1}{3} + 5 \cdot \frac{1}{3} + 6 \cdot \frac{1}{3} = 5.$$

Conditional expectation is useful in dividing complicated expectation calculations into simpler cases. We can find a desired expectation by calculating the conditional expectation in each simple case and averaging them, weighing each case by its probability.

For example, suppose that 49.6% of the people in the world are male and the rest female—which is more or less true. Also suppose the expected height of a randomly chosen male is 5' 11", while the expected height of a randomly chosen female is 5' 5." What is the expected height of a randomly chosen person? We can calculate this by averaging the heights of men and women. Namely, let H be the height (in feet) of a randomly chosen person, and let M be the event that the person is male and F the event that the person is female. Then

$$\begin{aligned} \text{Ex}[H] &= \text{Ex}[H \mid M] \Pr[M] + \text{Ex}[H \mid F] \Pr[F] \\ &= (5 + 11/12) \cdot 0.496 + (5 + 5/12) \cdot (1 - 0.496) \\ &= 5.6646 \dots \end{aligned}$$

which is a little less than 5' 8."

This method is justified by:

Theorem 18.4.5 (Law of Total Expectation). *Let R be a random variable on a sample space S , and suppose that $A_1, A_2, \dots$, is a partition of S . Then*

$$\text{Ex}[R] = \sum_i \text{Ex}[R \mid A_i] \Pr[A_i].$$